

An Ion Switch Regulates Fusion of Charged Membranes

Evgenios Siepi,[†] Silke Lutz,[†] Sylke Meyer,[‡] and Steffen Panzner^{†*}

[†]Novosom AG and [‡]Fraunhofer Center für Silizium-Photovoltaik, Halle, Germany

ABSTRACT Here we identify the recruitment of solvent ions to lipid membranes as the dominant regulator of lipid phase behavior. Our data demonstrate that binding of counterions to charged lipids promotes the formation of lamellar membranes, whereas their absence can induce fusion. The mechanism applies to anionic and cationic liposomes, as well as the recently introduced amphoteric liposomes. In the latter, an additional pH-dependent lipid salt formation between anionic and cationic lipids must occur, as indicated by the depletion of membrane-bound ions in a zone around pH 5. Amphoteric liposomes fuse under these conditions but form lamellar structures at both lower and higher pH values. The integration of these observations into the classic lipid shape theory yielded a quantitative link between lipid and solvent composition and the physical state of the lipid assembly. The key parameter of the new model, $\kappa(\text{pH})$, describes the membrane phase behavior of charged membranes in response to their ion loading in a quantitative way.

INTRODUCTION

Liposomes have recently regained much attention as carriers for oligonucleotides (ONs), such as antisense deoxynucleotides or siRNA. To be active, these large and highly charged molecules must be imported into the cytosol or the nucleus, and this process can be facilitated by liposomes (1,2). A fundamental problem with this import lies in the transition between the cargo-retaining state of the carrier outside the cell and the release of the encapsulated substance upon cellular contact. The low pH found in endosomes provides the trigger for such a transformation, and acid-induced fusion has been observed for lipid materials such as cholesterol hemisuccinate (CHEMS) and phosphatidylserine, in which the carboxyl function serves as the pH sensor (3–5). A major practical limitation of these anionic liposomes, however, is their limited ability to encapsulate ONs due to a lack of electrostatic interaction (6,7). Cationic liposomes can effectively sequester ONs, but they also display unspecific binding to serum components or endothelia (8–10). We recently demonstrated that both efficient loading of ON and high biocompatibility can be achieved with the use of amphoteric liposomes (11). These carriers adopt a cationic state at low pH but have anionic character at neutral pH. While investigating the pH-dependent fusion of amphoteric liposomes, we noticed the unexpected coexistence of two stable, lamellar phases that are observed at both low and neutral pH, with a fusogenic state that is limited to a zone around the isoelectric point of these membranes, typically about pH 5. This observation stands in contrast to previous studies that described a single, continuous transition between the lamellar and hexagonal phases of amphoteric membranes (12,13). This discrepancy prompted us to investigate the fusion mechanism of amphoteric liposomes. To that end, we first reexamined the phase

behavior of individual anionic or cationic species and then analyzed the more complex amphoteric assemblies.

MATERIALS AND METHODS

Morpholinocholesterol (MoChol) and cholesterylimidazol (CHIM) were synthesized as described previously (11,14).

Lipid structures, partial molecular volumes, and pK

The lipids used in this study, along with their abbreviations, structural formulas, partial molecular volumes, and ionization constants are listed in Table 1. We calculated the pK values by using the pK module of ACD/Labs 7.0 (Advanced Chemistry Development, Toronto, Canada) and further adjusted them by +1 or −0.5 for lipid anions or cations, respectively, to reflect the deviations between calculated and experimentally determined pK. For calculation purposes only, the pK of the ammonium group was set as 15 to merely reflect the constant charge of this moiety. Values with an underscore were used for the calculations.

All molecular volumes were determined with the use of DS Viewer Pro5.0 (Accelrys Software, San Diego, CA). For cholesterol, the coordinate file CLR of the Protein Data Bank (<http://www.rcsb.org/pdb>) was used to ensure the proper conformation of the molecule. The split point between the apolar and polar portions of each molecule was defined as the 3' carbon/oxygen bond for cholesterol derivatives, or as the C2–C3 bond for all diacylglycerols.

Molecular volumes for hydrated ions

The radii of the hydrated alkali ions were taken from Nightingale (15) and first converted into their space filling molecular volumes v_x (Table 2). We then calculated the hydration number n_H for these ions using the space filling volume for water of $29.9 \text{ \AA}^3$ (16). Further hydration numbers for acetate and chloride were obtained from Ohtaki and Radnai (17), and $n_H = 9$ was used for dihydrogen phosphate (17,18). Hydration numbers for the amino acids were obtained as the combined values of their side chains (19) and their zwitter ionic portion (20,21). For imidazole and tris (hydroxymethyl)aminomethane, we used the hydration numbers of the histidine and lysine side chains, respectively. The volumes for the central ions devoid of their coordinated waters were obtained from Nightingale (15) in the cases of alkali ions and chloride, and the van der Waals volumes for all other ions or water were determined using the DS viewer Pro5.0 software.

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*Correspondence: steffen.panzner@novosom.com

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TABLE 1 Description of lipids

Abbreviation and name	Structure	Head vol. [$\text{\AA}^3$] Tail vol. [$\text{\AA}^3$]	pK*
CHEMS		78.2	<u>5.41</u> 5.53 [†]
Cholesteryl hemisuccinate		343	5.8 [‡]
DOPA		62.8	7.38
1,2 Dioleoyl <i>sn</i> glycerol 3 phosphate		511.8	<u>2.83</u>
DOTAP		57.2	<u>15</u>
N [1 (2,3 Dioleoyloxy)propyl] N,N,N-trimethylammonium chloride		511.8	
CHIM		119.2	<u>6.36</u>
Cholesterol (3 imidazol 1 yl propyl) carbamate		343	6.0 [§]
MoChol		168.2	<u>6.51</u>
(α (3'O cholesteryloxycarbonyl) δ (N ethylmorpholine) succinamide)		343	6.50 [†]
POPG		115.9	<u>1.39</u>
1 Palmitoyl 2 oleoyl <i>sn</i> glycerol phosphatidylglycerol		490.4	

*For the calculation and adjustment of pK values, see [Materials and Methods](#); underlined values were used for the calculation of the phase diagrams.

[†]Experimentally determined for pure lipids according to Heyes et al. (28).

[‡]Reported in Hafez and Cullis (3) for CHEMS/POPC 1:1 mixtures.

[§]Reported in Budker et al. (14) for micellar solutions of CHIM.

The volumes of the hydrated ions v_{HI} were then calculated as $v_x + n_{\text{H}} \times v_{\text{H}_2\text{O}}$, where v_x is the crystal volume of the alkali ion and $v_{\text{H}_2\text{O}}$ is the volume of a water molecule.

TABLE 2 Molecular volumes for hydrated ions

Ion	r_x	r_h	v_x	v_x data	v_{app}	n_{H}	n_{H} data	v_{HI}
Li^+	0.60	3.82	1	(15)	233	7.8	calc	111
Na^+	0.95	3.58	4	(15)	192	6.3	calc	93
K^+	1.33	3.31	10	(15)	152	4.8	calc	77
Cl^-	1.81	3.32	25	(15)	153	4.3	calc	86
Ac			40	calc		5	(17)	111
H_2PO_4^-			48	calc		9	(17,18)	176
Glu			98	calc		12.5	(19,20)	275
Imid ⁺			51	calc		4	(19)	108
Tris ⁺			93	calc		4	(17,19)	150
Arg ⁺			131	calc		8	(19,20)	245

r_x and r_h are the crystal and hydrated radii of the ions, respectively, and v_x is the volume of the central ion. The data sources for these values are given in the " n_{H} data" column. The apparent molecular volume of the hydrated ion is v_{app} , and n_{H} is the hydration number as calculated from space filling volumes or according to data from the literature as indicated. The molecular volumes v_{HI} of the hydrated ions are also listed. All radii and volumes are in $\text{\AA}$ or $\text{\AA}^3$, respectively. Ac⁻, acetate; Glu⁻, glutamic acid; Imid⁺, imidazolium; Tris⁺, tris (hydroxymethyl)aminomethanium; Arg⁺, argininium.

Fusion after counterion discharge

Dowex 50WX2 was freshly prepared in its hydrogen form with the use of 1N hydrochloric acid, Dowex 1X2 was converted into its OH⁻ form with 1N sodium hydroxide, and both materials were extensively rinsed with water. Fluorescence resonance energy transfer (FRET) labeled liposomes were produced by injecting 320 μl of the 20 mM lipids in isopropanol into 10 ml of 50 mM acetic acid, 50 mM imidazole, pH 6. Both labeled species were combined and portions of the ion exchange materials were added. Aliquots were taken at pH 5, 4, and 3 upon addition of Dowex 50WX2 or at pH 7 or 8 upon addition of Dowex 1X2. Liposomes were incubated for 2 h at 37°C, after which the pH was adjusted back to 6 by means of acetic acid or imidazole, and the FRET signals were measured.

pH-dependent fusion of CHEMS and CHIM

CHEMS was dissolved in isopropanol at a concentration of 20 mM and supplied with 1 mol% NBD PE or N Rh PE, respectively, and liposomes were formed by injecting 100 μl of the lipid solution into 700 μl of buffer A (10 mM acetic acid and phosphoric acid, pH 7.5 adjusted with NaOH). Then 100 μl of NBD PE labeled and 100 μl of N Rh PE labeled liposomes were mixed together with 200 μl buffer A. Aliquots of 50 μl were brought to the indicated pH using 50 μl of 5 $\times$ buffer A adjusted to the target pH values. Fusion was allowed for 2 h at 37°C, after which the suspensions were neutralized to pH 7.5 using 50 μl of appropriately concentrated NaOH to more clearly distinguish between liposome aggregation and fusion.

Fluorescence was recorded after completion of the pH cycle. Liposomes from CHIM were produced from 20 mM CHIM in isopropanol as described above, and fusion of CHIM in the presence of chloride ions was investigated in buffers consisting of 10 mM each of L lysine, pyridine, and imidazole, aliquots of which were adjusted to the respective pH values with hydrochloric acid. Individually labeled liposomes from CHIM were produced at pH 4 and brought to the indicated pH by 5× buffer. Fusion was allowed for 2 h and fluorescence was recorded after readjustment to pH 3.8.

Fusion of amphoteric liposomes in response to pH

Liposomes were prepared by injecting lipid mixtures in isopropanol (20 mM) into buffer A to a final concentration of 3 mM. Then 100 μ l of NBD PE labeled and 100 μ l of N Rh PE labeled liposomes of otherwise identical composition were mixed together with 200 μ l buffer A. Aliquots of 50 μ l were adjusted to the indicated pH using 50 μ l of 5× buffer A adjusted to the respective pH. Fusion was allowed for 2 h at 37°C, after which FRET signals were recorded. To discriminate between fusion and mere aggregation, we then neutralized the suspensions to pH 7.5 using 50 μ l of appropriately concentrated NaOH and again recorded the fluorescence. All liquid handling was performed with the use of a Multiprobe II Ex robot (Perkin Elmer, Waltham, MA) in black 96 well plates. The presence of residual amounts of isopropanol did not result in any appreciable change in the fusion properties.

Ion binding

For ion binding, 300 μ l of 30 mM lipid solutions in isopropanol were injected into 2 ml of 10 mM sodium phosphate, pH 7.5. Liposomes were separated from solvent ions using PD10 columns (GE, Uppsala, Sweden) equilibrated in 15% isopropanol/water. Lipid recovery was ~80%. Liposome bound sodium and phosphorus concentrations were determined with the use of an Element 2 inductively coupled plasma mass spectrometer (Thermo Scientific, Bremen, Germany) as described previously (22).

Further details regarding mathematical considerations and the materials and methods used are presented in the [Supporting Material](#).

RESULTS

Membrane fusion is regulated by counterion binding

The stabilization of pH-sensitive bilayers (e.g., from CHEMS) has been explained by the hydration of the charged headgroup, the electrostatic repulsion between these moieties, through binding of counterions or a combination of these elements (3,13). To discriminate between these assumptions, we analyzed the fusion of charged and uncharged bilayers of CHEMS under conditions of counterion binding or dissociation. We monitored fusion through lipid mixing between individually labeled membranes using FRET. For this purpose, particles were prepared and mixed at neutral pH, incubated at various lower values of pH, and eventually neutralized to discriminate fusion from mere agglomeration.

Liposomes from anionic CHEMS ($pK \sim 5.4$; [Table 1](#)) were prepared at pH 6 in the presence of charged imidazolium ions ($pK = 7.0$ (23)). The lipid particles were stable and did not fuse, as would be expected from the theories mentioned above.

We then discharged the imidazolium ions by adding small portions of the anion exchange resin DOWEX 1X2 in its OH form. This technique liberates hydroxyl ions and raises the pH, but avoids the addition of interfering cations. The procedure resulted in the fusion of CHEMS membranes at neutral or alkaline pH. When the pH of the CHEMS/imidazole system was raised by the addition of sodium hydroxide, no fusion was observed ([Fig. 1, A and B](#)). This difference in the experimental outcome deemphasizes electrostatic repulsion or headgroup hydration as mechanisms stabilizing the

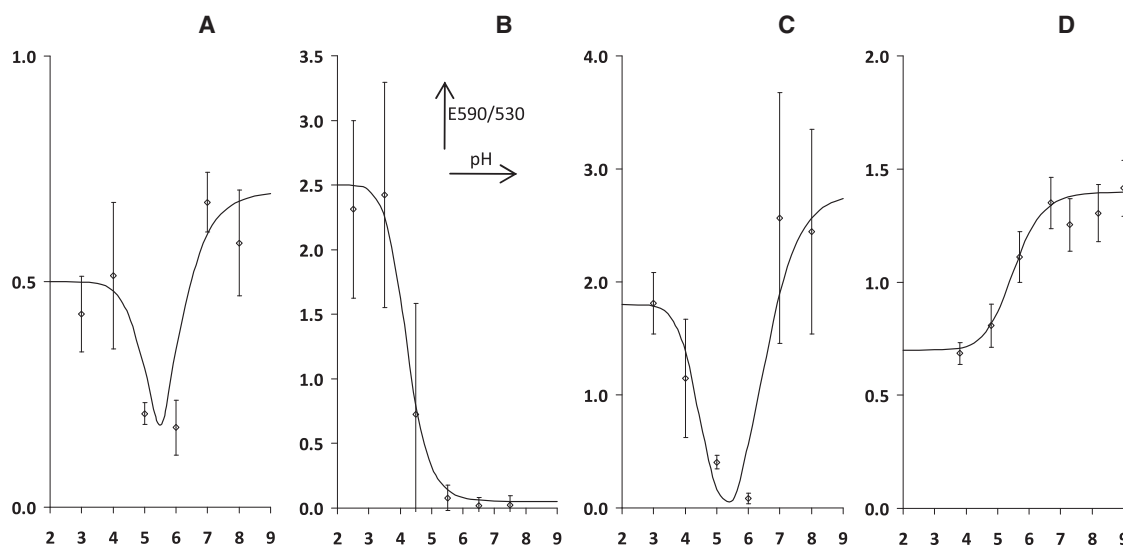


FIGURE 1 Counterion dependent membrane fusion. (A) FRET labeled liposomes from CHEMS were formed in imidazole/acetate buffer at pH 6, and the pH was adjusted by addition of ion exchange materials in their H^+ or OH^- form. Fusion of the lipid materials is observed at both high and low pH. (B) Liposomes as in A were produced in the presence of Na^+ and the pH was adjusted by addition of acid or base. Fusion at low pH occurs as in A, but no fusion is observed at high pH. (C) FRET labeled liposomes from CHIM were formed and ion exchange materials were added as in A. Fusion of the liposomes is observed at both high and low pH. (D) Liposomes as in C were formed in lysine/morpholine/imidazole buffer at pH 4. Stronger fusion is observed at high pH when CHIM becomes discharged.

lamellar phase, as these relate to the charge status of CHEMS, a variable that is unchanged in the experiment. Instead, our data identify ion decoration as a critical component. Consistent with this finding, protonation of CHEMS at low pH resulted in fusion regardless of whether this was achieved by addition of hydrochloric acid or the use of H^+ loaded ion exchange materials. In both cases, CHEMS lost its ion-binding capacity through protonation of its polar headgroup.

Liposomes made from a cationic, pH-sensitive cholesteryl-imidazol (CHIM (14), $pK \sim 6.4$) showed a reciprocal behavior. Fusion was not observed between membranes of CHIM in the presence of acetate ions at pH 6, but was induced upon addition of the cation exchanger DOWEX 50WX2 in its H^+ form. As protons are released from the cation exchanger, acetate ions are neutralized, which in turn can lead to their dissociation from the membrane. In contrast, no phase transition was achieved upon direct acidification with hydrochloric acid. Because CHIM is positively charged regardless of the manner in which acidification of the medium is achieved, the difference in its phase behavior most likely relates to the recruitment of the counterions from solvent. The chloride ions introduced with HCl apparently bind to the imidazole headgroups of CHIM, thereby stabilizing the lamellar state of the membrane, whereas acidification by DOWEX 50WX2 does not introduce the stabilizing chloride ions, and fusion is observed. Fusion of CHIM liposomes at higher pH was observed both upon addition of OH^- loaded ion exchange materials and upon adjustment of the pH with sodium hydroxide (Fig. 1, C and D). In both cases, CHIM lost its cationic charge and therefore its ability to bind solvent ions. As with CHEMS, fusion of CHIM coincided with ion recruitment to the bilayer, but was not linked to headgroup repulsion or hydration.

Ion size relates inversely to fusion

Because ion binding coincides with the appearance of a non-fusogenic state of the membrane, we were interested in determining whether the size of the bound ions can modulate fusion. To test this notion, we prepared FRET-labeled liposomes from CHEMS and cholesterol (15:85 mol%), and monitored lipid mixing over time. Decreased lipid mixing was observed in the series of $K^+ > Na^+ > Li^+ > \text{tris(hydroxymethylaminomethan)}^+ > \text{arginine}^+$. Control reactions using anions of different sizes in combination with Na^+ did not result in any notable differences in the fusion process (Fig. 2 A). Conversely, the fusion of cationic liposomes made from CHIM/cholesterol (20:80 mol %) was enhanced in the order of glutamate < acetate < chloride, but was unchanged in acetate buffers comprised of K^+ , Na^+ , or Li^+ cations (Fig. 2 B).

The fusogenicity of a charged membrane therefore depends on the size of the attracted ions, and large counter-

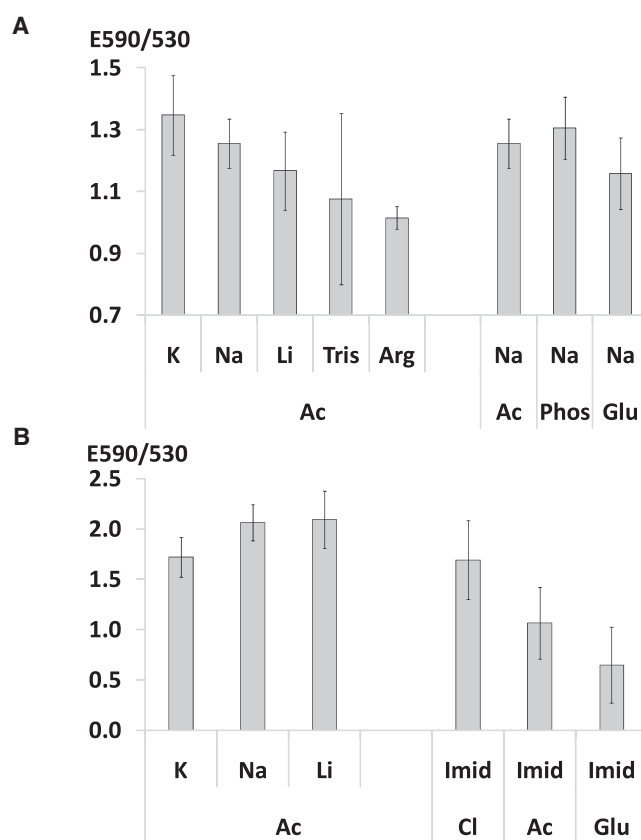


FIGURE 2 Counterion size modulates lipid membrane fusion. FRET labeled liposomes from CHEMS/Chol (A, 15:85) or CHIM/Chol (B, 20:80) were formed at pH 6 in buffers containing the indicated ions, and lipid mixing between vesicles was monitored over time. (A) The extent of fusion follows the volumes v_{HI} of the hydrated cations (see Table 2), whereas various anions do not affect the fusion properties of CHEMS/Chol. (B) Conversely, the large anion glutamic acid (Glu), but not the smaller chloride, suppresses fusion of CHIM/Chol. Different cations have no impact on fusion.

ions interfere more strongly with membrane fusion than do smaller ones. The ions in water exist as hydrated species, and we took this into account for the molecular volumes presented in Table 2.

Taken together, our results support an ion switch model for membrane fusion. In this model, the presence of lipid-bound solvent ions promotes membrane stability, whereas their absence can lead to fusion.

Amphoteric liposomes show a double phase transition

Mixtures of anionic and cationic lipids can form amphoteric liposomes provided that at least one of the components is pH-sensitive. On the basis of the above analysis, we expected the amphoteric membranes to have little or no phase transition, because the membrane-stabilizing ion binding of the anionic lipid should complement that of the cationic amphiphile. Alternatively, amphoteric liposomes formed

from a constantly charged cationic lipid in combination with CHEMS should display a dampened but continuous phase transition, as described by Hafez et al. (12). In contrast to that earlier report, we observed lipid fusion at pH 6 and pH 4.5, but the existence of lamellar structures at both pH 7.5 and pH 3 for amphoteric liposomes from CHEMS and dioleoyl-(trimethylammonium)propanediol (DOTAP; Fig. 3).

This observation prompted us to systematically probe the pH-induced fusion for binary mixtures of CHEMS and DOTAP. For amphoteric systems (i.e., mixtures with an excess of CHEMS), reduced lipid mixing and maintenance of particle size were observed at both neutral and low pH, but fusion occurred around pH 5. This double phase transition is unique to amphoteric mixtures and was not observed in cationic blends with an excess of DOTAP (Fig. 4, A and B). Vesicles that were rich in DOTAP displayed a size increment but did not fuse. We attribute this to a cross-linking of DOTAP liposomes in the presence of the bivalent phosphate ions, because the effect disappeared in the presence of monovalent buffers.

In accordance with our initial expectation, and given the nonfusogenic character of DOTAP (24), the addition of DOTAP should have dampened the acid-induced fusion of CHEMS. Instead, we observed fusion at slightly acidic conditions and the existence of two lamellar phases at both neutral and acidic pH in amphoteric mixtures of the oppositely charged lipids. This can be explained by the formation

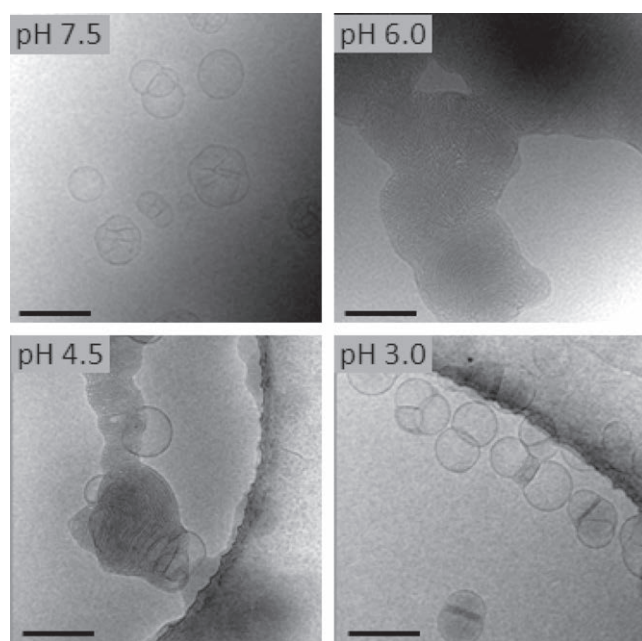


FIGURE 3 Amphoteric liposomes display bistable phase behavior. Liposomes from DOTAP and CHEMS (45:55) were produced at pH 7.5, exposed to the pH indicated, and examined by cryo transmission electron microscopy. The material forms a lamellar phase at both pH 7.5 and pH 3, but undergoes a phase transition at pH 6 or pH 4.5. Bars = 200 nm.

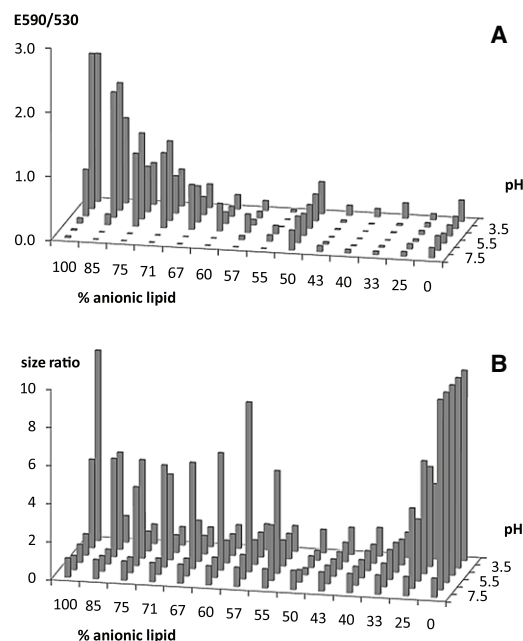


FIGURE 4 pH induced fusion of binary mixtures from DOTAP and CHEMS. Systematically varied blends of DOTAP and CHEMS were dissolved in isopropanol, split, and labeled with FRET marker lipids. After formation of individually labeled liposomes was completed, matching samples were recombined, the materials were exposed to more acidic pH for 2 h, and then readjusted to neutrality. pH induced fusion was monitored by the appearance of the FRET signal (A) or the formation of larger particles (B). The size increment in B denotes the ratio of the particle sizes before and after the pH cycle.

of an intrabilayer lipid salt, a structure that is fusion-promoting because it is devoid of solvent counterions.

DOTAP and CHEMS form a lipid salt devoid of counterions

To test the occupancy of mixtures from DOTAP and CHEMS with solvent ions, we generated liposomes in sodium phosphate buffer at pH 7.5 and separated unbound ions by gel filtration in pure water. The lipid-bound sodium or phosphorus was then quantified by inductively coupled plasma mass spectroscopy (ICP-MS). Apart from low amounts of passively trapped sodium or phosphate, we measured low levels of bound solvent ions in samples with nearly equal amounts of DOTAP and CHEMS (Fig. 5). Liposomes with an excess of CHEMS did adsorb sodium but showed only background levels of phosphorus, whereas those with an excess of DOTAP recruited phosphate almost exclusively. Eventually, the adsorbed amounts were in proportion to the excess of the respective lipid, not to its total amount. Taken together, these facts support the formation of an ion-free lipid salt within mixed bilayers of DOTAP and CHEMS. For the remainder of the free, unpartnered lipid, our data provide direct experimental evidence for a recruitment of solvent ions to charged lipid species.

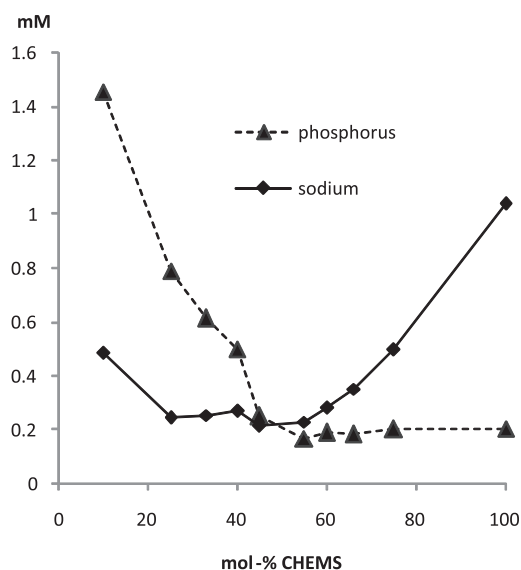


FIGURE 5 Ion adsorption to lipid membranes. Liposomes from various mixtures of DOTAP and CHEMS were produced in sodium phosphate buffer and separated from unbound solvent ions through size exclusion chromatography in water. Bound sodium and phosphorus were measured by ICP MS. The mixed membranes from DOTAP and CHEMS bind sodium whenever CHEMS is present in excess, but they bind phosphorus in the presence of excess DOTAP. Quantitatively, the ion binding capacity of these membranes follows excess of either lipid, which is the material that does not participate in the lipid salt. The ion binding reaches a minimum for equimolar mixtures of DOTAP and CHEMS.

Quantitative modeling

We next set out to quantify our observations using the framework of the lipid shape theory. This model relates membrane fusion to a low aspect ratio between the polar and apolar regions of a lipid and the formation of a lamellar phase to higher ratios (25). The important role played by ions in membrane fusion and stabilization required an extension of the classic model. Here, we include ions as volume-contributing elements of the lipid structures. In the example given, binding of a hydrated sodium ion adds $93 \text{ \AA}^3$ to the volume of the hemisuccinate portion of CHEMS that by itself occupies only $78 \text{ \AA}^3$. Conversely, a hydrated dihydrogen phosphate ion adds $176 \text{ \AA}^3$, respectively, to the headgroup of DOTAP, which itself has a volume of only $57 \text{ \AA}^3$. The volume contribution made by a bound ion is therefore substantial.

The general function for the pH-dependent phase behavior of a charged bilayer can now be written as

$$\kappa(\text{pH}) = \frac{(x * V_{\text{AH}} + (1 - x) * V_{\text{CH}} + x_{\text{I}} * V_{\text{I}})}{(x * V_{\text{AT}} + (1 - x) * V_{\text{CT}})}, \quad (1)$$

where κ describes the volume ratio between all polar and apolar elements of the bilayer, and x is the molar fraction of the anionic lipid. The polar elements comprise the headgroup volumes V_{AH} and V_{CH} of the lipid anion and cation,

and an amount x_{I} of the respective counterions with a volume V_{I} , the latter being equal to the fraction of charged but unpaired lipid headgroups:

$$\begin{aligned} x_{\text{I}} &= |x_{\text{A}^-} - x_{\text{C}^+}| \\ &= x / (1 + c_{\text{H}^+} / K_{\text{A}}) - (1 - x) / (1 + K_{\text{C}} / c_{\text{H}^+}) \end{aligned}$$

where K_{A} and K_{C} are the ionization constants of the lipid anion and cation, respectively, and c_{H^+} is the proton concentration. The apolar volume elements are contributed by the tail volumes V_{AT} and V_{CT} , respectively.

For membranes that comprise both anionic and cationic lipids, $\kappa(\text{pH})$ describes the stabilization of a lipid membrane through attraction of a counterion volume V_{I} to the portion x_{I} of charged lipid molecules that do not form the lipid salt. First, low values for κ indicate the fusogenic state of the lipid assembly that relates to a relatively small headgroup volume, whereas higher values are linked to the formation of a lamellar phase. When calculated for amphoteric mixtures of CHEMS and DOTAP, $\kappa(\text{pH})$ reflects the stable lipid phases observed at both low and high pH. Second, a fusogenic phase around the isoelectric point appears both in the experiment and in $\kappa(\text{pH})$. Third, $\kappa(\text{pH})$ follows the single-sided, monophasic pH dependency for CHEMS. Fourth, the calculation predicts the non-fusogenic bilayers observed for mixtures that have an excess of DOTAP or are formed from pure DOTAP. Fifth, low values for $\kappa(\text{pH})$ predict a fusogenic state for the equimolar mixture of DOTAP and CHEMS (Fig. 6). In fact, we observed lipid mixing for this composition at neutral pH.

Biphasic stability is a general feature of amphoteric liposomes

Expanding on these results, we also analyzed amphoteric liposomes in which both the anionic and cationic lipids are pH-sensitive. We classify these systems as amphoteric II, in

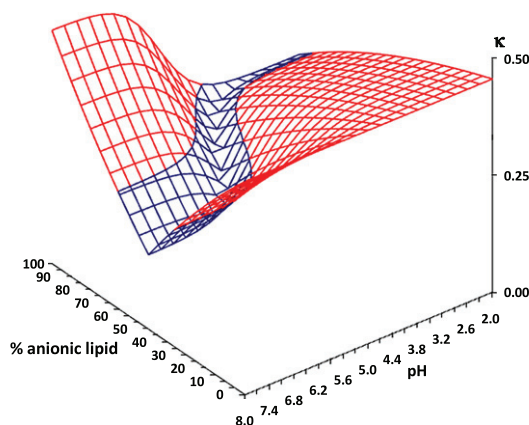


FIGURE 6 Calculated phase diagram for DOTAP/CHEMS using the assumptions of counterion binding and the formation of the ion free lipid salt. The volume ratio κ assumes a pH dependent minimum in the amphoteric mixtures with >50% CHEMS, the appearance of which correlates with the fusion zone observed in Figs. 3 and 4.

contrast to amphoter I, which consists of a weak anionic amphiphile in combination with a strong lipid cation.

Indeed, amphoter II liposomes constructed from either CHIM/CHEMS or analogous systems composed of MoChol (11) and CHEMS also show biphasic stability and fusion around their respective isoelectric points (Fig. 7, A and B). If MoChol is used as the cationic component, this behavior is limited to mixtures with an excess of CHEMS, whereas a composition rich in MoChol no longer undergoes pH-induced fusion. We attribute this to the rather large polar headgroup of MoChol, which suppresses fusion by contributing volume to this portion of the molecule, quite analogously to the binding of very large counterions. Calculations according to Eq. 1 reflect the lipid phase behavior of both amphoter II systems in detail: low values for $\kappa(\text{pH})$ coincide with the occurrence of lipid mixing, whereas higher values for $\kappa(\text{pH})$ indicate the formation of a lamellar phase (Fig. 8, A and B).

Biphasic stability was also observed for an amphoter III system comprised of the stably charged anionic lipid dioleoylphosphatidic acid (DOPA) and an excess of the pH-dependent cationic lipid MoChol, as reflected in the

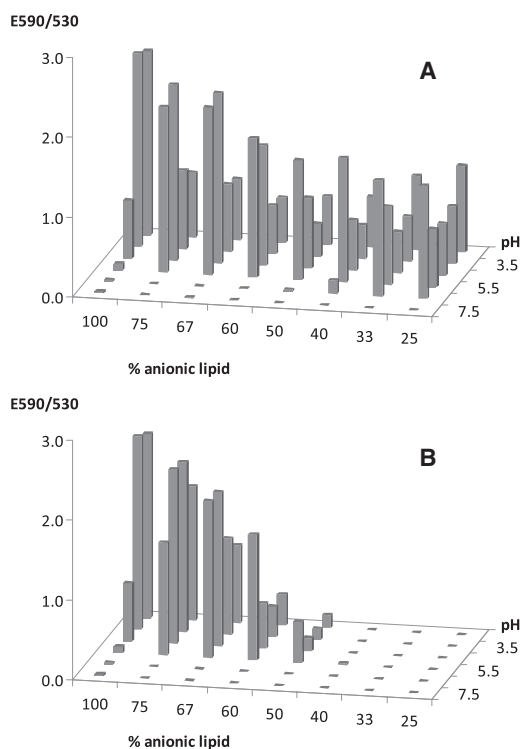


FIGURE 7 The pH dependent fusion properties of amphoter II systems. Liposomes containing the indicated amounts of anionic lipid were produced from CHIM and CHEMS (A) or MoChol and CHEMS (B), and their pH dependent lipid mixing was monitored by FRET. All mixtures of CHIM/CHEMS have amphoteric character and display pH dependent fusion. Mixtures of MoChol/CHEMS are also all amphoteric, but liposomes with high amounts of MoChol do not fuse, probably due to the larger volume of the MoChol headgroup.

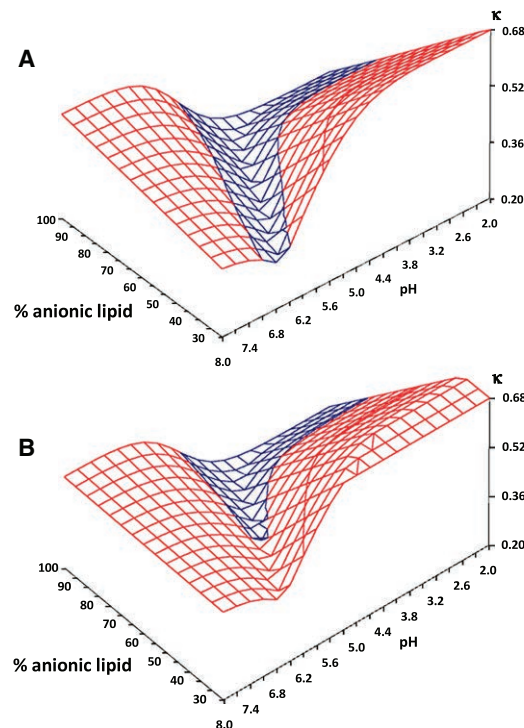


FIGURE 8 Phase diagrams for amphoter II systems. The phase behavior of CHIM/CHEMS (A) and MoChol/CHEMS (B) was calculated assuming the formation of an ion free lipid salt and ion recruitment to the overage of charged lipids. Higher values for $\kappa(\text{pH})$ reflect the existence of a stable lipid phase at acidic and neutral pH, whereas lower values correlate with the existence of a fusogenic phase at slightly acidic conditions. (B) The inhibition of fusion for mixtures with >50 mol % of MoChol corresponds with high values of $\kappa(\text{pH})$ in the phase diagram.

respective function $\kappa(\text{pH})$ shown in Fig. 9. For these calculations, we used the monovalent form of DOPA because that is the prevalent form of the molecule below pH 7, a condition that is required for the protonation of MoChol. However, one should keep in mind that DOPA acquires an additional charge above pH 7 and may bind a second counterion, possibly leading to a further increase of its headgroup size. An interesting exception to the general picture presented here is the absence of fusion in MoChol/POPG, which may be caused by steric hindrance (Fig. S2).

DISCUSSION

This work features lipid-bound ions as a regulator of membrane fusogenicity. In this model, ion adsorption stabilizes charged membranes, whereas ion desorption can lead to fusion. We directly demonstrated this by using the novel approach of counterion discharge, which enables the preparation of charged but ion-depleted lipid membranes. Charged membranes in the presence of discharged counterions undergo fusion, as evidenced by lipid mixing from differently labeled membranes. Because the charge of the polar lipid headgroup remains constant during the

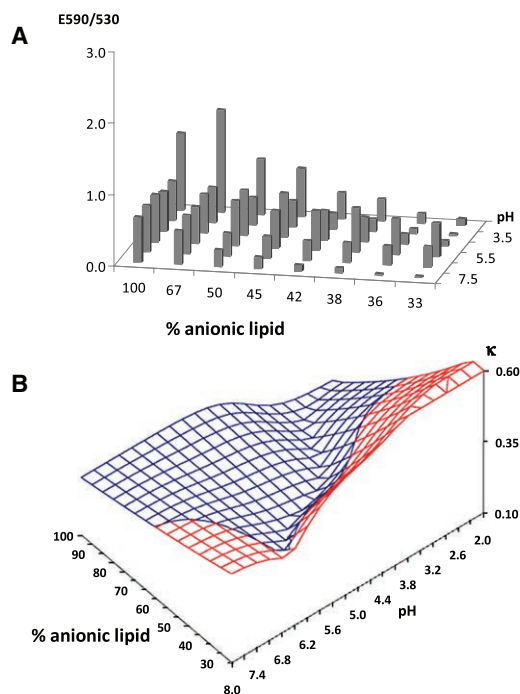


FIGURE 9 A pH dependent fusion and phase diagram of an amphoteric III system. Liposomes containing the indicated amounts of anionic lipid were produced from MoChol and DOPA (A) and their pH dependent lipid mixing was monitored by FRET. Mixtures with 33/50 mol % of DOPA have amphoteric character and display pH dependent fusion. (B) Phase diagram for MoChol/DOPA assuming a lipid salt formation and counterion recruitment.

counterion discharge, the observed phase transition cannot be explained by changes in the electrostatic repulsion between individual lipids or variations of lipid headgroup hydration. The fusogenicity of these systems was even sufficient to overcome the electrostatic repulsion between lipid particles. This leaves counterion binding as the most direct explanation for the experimental observations, and the reduced membrane fusion in the presence of larger counterions suggests that the volume rather than the chemistry of an ion is important. Ion recruitment to charged bilayers is sufficient to explain the well-described fusion of CHEMS-liposomes upon acidification (3), and it also reflects the fusion of CHIM-liposomes at higher pH (Fig. S1 and Fig. S3). Recent molecular-dynamics simulations confirmed the recruitment of solvent ions to bilayers of CHEMS in response to the charge of the lipid (26).

The pH-related stabilization of charged lipid membranes occurs in a reciprocal fashion for anionic or cationic lipids. Therefore, one might expect mixed membranes to be stable over the entire range of pH conditions. This was clearly not the case, however, as demonstrated for various amphoteric liposomes constructed from weak lipid anions and strong cations (amphoteric I), weak lipid anions and weak cationic amphiphiles (amphoteric II), or mixtures of strong lipid anions and weak cationic amphiphiles (amphoteric III).

Instead, most of these systems underwent fusion around their isoelectric point. This can be explained by the formation of a lipid salt, whereby the two oppositely charged amphiphiles neutralize each other. Lipid salt formation leads to a displacement of counterions and a concomitant reduction in the headgroup volume, which in turn causes fusion. Experimental evidence for the hypothesized lipid salt formation was provided by a quantitative analysis of the membrane-bound solvent ions in binary mixtures of DOTAP and CHEMS, which showed minimized ion decoration under conditions of charge neutralization. In addition, we observed ion selectivity for mixtures with a modest excess of CHEMS or DOTAP. This observation is best explained by the sequestration of the minor lipid component in a lipid salt, such that only the lipid in excess is available for counterion recruitment and membrane stabilization. In systems with an excess of CHEMS, sodium recruitment is sufficient to stabilize the lamellar phase of the DOTAP/CHEMS membrane at neutral pH. Acidification reduces the amount of charged CHEMS in the system until at the isoelectric point its molar fraction is equal to that of DOTAP. The membrane is now ion-free and highly fusogenic because all of the charged CHEMS forms a lipid salt with DOTAP and the remainder of the anionic lipid is protonated. Further acidification produces more uncharged CHEMS, which leads to a liberation of DOTAP from the lipid salt, recruitment of acetate or phosphate, and concomitant membrane stabilization (Fig. S3).

Taken together, our results indicate that the binding of solvent ions to charged lipid headgroups is a dominant regulator of the phase behavior of lipid membranes. The process provides a universal explanation for the seemingly different fusion processes observed in charged but ion-depleted systems or discharged membranes. In combination with the formation of a lipid salt, it also fully explains the complex phase behavior of amphoteric membranes.

The important role of lipid-bound ions led us to extend the classic lipid shape theory, which relates lipid geometry to phase behavior. We now introduce the lipid-bound ion as an integral component of the lipid headgroup that contributes to its volume. This novel (to our knowledge) dynamic shape theory reflects the pH-dependent ion recruitment to lipid bilayers and describes the lipid phase behavior in context with a solvent. Our model calculations reflect the complex phase behavior observed in experiments, and yield quantitative results within a given system or for related systems. We noticed, though, that the fusion of DOTAP/CHEMS starts at lower values of κ compared with that of amphoteric II or III systems. One might relate this to the nature of the hydrophobic tail regions; however, both the amphoteric I and amphoteric III systems represent a combination of a diacylglycerol and a sterol, whereas the amphoteric II systems are entirely sterol-based. Thus, the split point between the apolar and polar lipid fragments remains a sensitive variable in any lipid shape calculations.

A second variable is the actual volume contribution of the counterions, as these exist as hydrated species in water and their hydration numbers n_H may vary with the method of their determination (17) or during salt formation or complexation (27). In our calculations, we consider the n_H of a charged lipid headgroup in complex with its counterion to be equal to that of the hydrated solvent ion. This assumption can account for the partial reduction of the hydration shells of both the lipid and the solvent ion during complexation. However, the uncertainty related to n_H is modest, because a Δn_H of 2 results in an inaccuracy of $\sim 15\%$ when calculated for sodium hemisuccinate. Of more importance, assumptions for n_H never affect the qualitative outcome of the model they only change the degree of stabilization achieved by ion binding.

CONCLUSIONS

In summary, in this work we identified membrane-bound ions, or an ion switch, as a key regulator of the stability of charged membranes. We used this initial discovery, together with the hypothesis regarding lipid salt formation, to explain the double phase transitions of amphoteric liposomes. Eventually, our observations led to a dynamic shape theory that describes the membrane in context with solvent ions. Our future work will demonstrate the applicability of this theory for multicomponent systems involving neutral lipids and shall eventually provide a structure-activity relationship linking liposome composition and performance in cell transfection.

SUPPORTING MATERIAL

Additional materials and methods, the detailed mathematical background for the dynamic shape theory, supporting experimental data, and a model scheme are available at [http://www.biophysj.org/biophysj/supplemental/S00063495\(11\)004061](http://www.biophysj.org/biophysj/supplemental/S00063495(11)004061).

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